

LABORATORY WORK 2: DATABASE NORMALIZATION

Task 1: Violated Normal Forms

Unnormalized Table: `Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)`

- **1NF Violation:** Column `seat_numbers` contains non-atomic comma-separated values (e.g., "12A, 12B").
- **2NF Violation:** Partial dependencies exist where non-key attributes (e.g., `flight_number`, `ticket_price`) depend only on `booking_id`, not the entire composite key (`booking_id`, `passenger_passport_number`).
- **3NF Violation:** Transitive dependencies exist between non-key attributes (`departure_airport_name` → `departure_city`, `flight_number` → `airline_name`).

Task 2: Step-by-Step Normalization Process

Step 1: 1NF (Atomic Values)

- **Action:** Split multi-value seats into individual rows. Primary Key: (`booking_id`, `passenger_passport_number`).
- **1NF_Booking_receipt** (`booking_id`, `passenger_passport_number`, `passenger_full_name`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name`, `seat_number`, `ticket_price`)

Step 2: 2NF (Remove Partial Dependencies)

- **Action:** Extract attributes depending only on `booking_id` or `passenger_passport_number` into separate tables.
1. **Passengers** (`passenger_passport_number`, `passenger_full_name`)
 2. **Bookings** (`booking_id`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name`, `ticket_price`)
 3. **Booking_Passengers** (`booking_id`*, `passenger_passport_number`*, `seat_number`)

Step 3: 3NF (Remove Transitive Dependencies)

- **Action:** Extract indirect dependencies ($\text{airport} \rightarrow \text{city}$, $\text{flight} \rightarrow \text{airline/airports}$).
1. **Airports** (**airport_name**, city)
 2. **Airlines** (**airline_name**)
 3. **Flights_3NF** (**flight_number**, departure_airport_name*, arrival_airport_name*, airline_name*)
 4. **Bookings_3NF** (**booking_id**, flight_number*, ticket_price)
 5. **Passengers_3NF** (**passenger_passport_number**, passenger_full_name)
 6. **Booking_Passengers** (**booking_id***, **passenger_passport_number***, seat_number)

Task 3: Functional Dependencies (FDs)

- **FD1:** $\text{booking_id} \rightarrow \text{flight_number}, \text{ticket_price}$
- **FD2:** $(\text{booking_id}, \text{passenger_passport_number}) \rightarrow \text{seat_number}$
- **FD3:** $\text{passenger_passport_number} \rightarrow \text{passenger_full_name}$
- **FD4:** $\text{flight_number} \rightarrow \text{departure_airport_name}, \text{arrival_airport_name}, \text{airline_name}$
- **FD5:** $\text{departure_airport_name} \rightarrow \text{departure_city}$
- **FD6:** $\text{arrival_airport_name} \rightarrow \text{arrival_city}$

Task 4: DDL and Data Population for 3NF Tables

SQL

-- 1. Create Tables

```
CREATE TABLE Airports (
    airport_name VARCHAR(100) PRIMARY KEY,
    city VARCHAR(100) NOT NULL
);
```

```
CREATE TABLE Airlines (
    airline_name VARCHAR(100) PRIMARY KEY
);
```

```
CREATE TABLE Flights_3NF (
    flight_number VARCHAR(20) PRIMARY KEY,
    departure_airport_name VARCHAR(100) NOT NULL REFERENCES
    Airports(airport_name),
    arrival_airport_name VARCHAR(100) NOT NULL REFERENCES Airports(airport_name),
    airline_name VARCHAR(100) NOT NULL REFERENCES Airlines(airline_name)
);
```

```
CREATE TABLE Bookings_3NF (
    booking_id INT PRIMARY KEY,
    flight_number VARCHAR(20) NOT NULL REFERENCES Flights_3NF(flight_number),
    ticket_price DECIMAL(10, 2) NOT NULL
);
```

```
);
```

```
CREATE TABLE Passengers_3NF (  
    passenger_passport_number VARCHAR(20) PRIMARY KEY,  
    passenger_full_name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Booking_Passengers (  
    booking_id INT NOT NULL REFERENCES Bookings_3NF(booking_id),  
    passenger_passport_number VARCHAR(20) NOT NULL REFERENCES  
Passengers_3NF(passenger_passport_number),  
    seat_number VARCHAR(10) NOT NULL,  
    PRIMARY KEY (booking_id, passenger_passport_number)  
);
```

-- 2. Insert 5 Sample Rows

```
INSERT INTO Airports VALUES ('JFK', 'New York'), ('LHR', 'London'), ('ALA', 'Almaty'),  
('TSE', 'Astana'), ('DXB', 'Dubai');
```

```
INSERT INTO Airlines VALUES ('Air Astana'), ('Emirates'), ('British Airways'), ('Delta'),  
('FlyArystan');
```

```
INSERT INTO Flights_3NF VALUES ('KC875', 'ALA', 'TSE', 'Air Astana'), ('EK518', 'DXB',  
'ALA', 'Emirates'), ('BA117', 'LHR', 'JFK', 'British Airways'), ('DL401', 'JFK', 'LHR', 'Delta'),  
('KC951', 'ALA', 'DXB', 'Air Astana');
```

```
INSERT INTO Bookings_3NF VALUES (101, 'KC875', 150.00), (102, 'EK518', 450.00), (103,  
'BA117', 600.00), (104, 'DL401', 550.00), (105, 'KC951', 300.00);
```

```
INSERT INTO Passengers_3NF VALUES ('N123456', 'John Doe'), ('N654321', 'Jane Smith'),  
('N987654', 'Aliya Sarina'), ('N456789', 'Arman Murat'), ('N112233', 'Elena Rostova');
```

```
INSERT INTO Booking_Passengers VALUES (101, 'N123456', '12A'), (101, 'N654321',  
'12B'), (102, 'N987654', '05C'), (103, 'N456789', '14F'), (104, 'N112233', '22D');
```

Task 5: 3NF Verification for **Flights** Table

Question: Does the **Flights** table in Part 2 satisfy 3NF? **Answer: Yes, it is in 3NF.**

Reasoning: All non-key attributes depend directly and only on the primary key **flight_id**. Transitive attributes like airport details and airline details are removed and referenced via foreign keys (**departing_airport_id**, **arriving_airport_id**, **airline_id**). There are no non-key functional dependencies remaining inside the table.